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ROLL NO:-2520030188

EXP 1:-

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}
```

OUTPUT:-

```
praneeth@PRANEETH:~$ nano 1.c
praneeth@PRANEETH:~$ gcc 1.c -o 1
praneeth@PRANEETH:~$ ./1
Enter Linux command: uname
Parent PID : 1129
Child PID : 1130
Linux
Child process completed.
praneeth@PRANEETH:~$ ./1
Enter Linux command: lscpu
Parent PID : 1140
Child PID : 1141
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                39 bits physical, 48 bits virtual
Byte Order:                   Little Endian
CPU(s):                       12
On-line CPU(s) list:         0-11
Vendor ID:                    GenuineIntel
Model name:                   13th Gen Intel(R) Core(TM) i5-13420H
CPU family:                   6
Model:                        186
Thread(s) per core:          2
Core(s) per socket:          6
Socket(s):                    1
Stepping:                     2
BogoMIPS:                     5222.40
Flags:                         fpu vme de pse tsc msr pae mce cx8 apic sep mtr
                               r pge mca cmov pat pse36 clflush mmx fxsr sse s
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r pge mca cmov pat pse36 clflush mmx fxsr sse s
se2 ss ht syscall nx pdpe1gb rdtscp lm constant
_tsc rep_good nopl xtopology tsc_reliable nonst
op_tsc cpuid tsc_known_freq pni pclmulqdq vmx s
sse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe p
opcnt tsc_deadline_timer aes xsave avx f16c rdr
and hypervisor lahf_lm abm 3dnowprefetch ssbd i
brs ibpb stibp ibrs_enhanced tpr_shadow ept vpi
d ept_ad fsgsbase tsc_adjust bml avx2 smep bmi
2 erms invpcid rdseed adx smap clflushopt clwb
sha_ni xsaveopt xsavec xgetbv1 xsaves avx_vnni
vnm i umip waitpkg gfni vaes vpclmulqdq rdpid mo
vdiri movdir64b fsrm md_clear serialize ibt flu
sh_lli arch_capabilities

Virtualization features:
  Virtualization: VT-x
  Hypervisor vendor: Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d: 288 KiB (6 instances)
  L1i: 192 KiB (6 instances)
  L2: 7.5 MiB (6 instances)
  L3: 12 MiB (1 instance)
NUMA:
  NUMA node(s): 1
  NUMA node0 CPU(s): 0-11
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite: Not affected
  Indirect target selection: Not affected
  Itlb multihit: Not affected
  L1tf: Not affected
  Mds: Not affected
  Meltdown: Not affected
  Mmio stale data: Not affected
  Old microcode: Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed: Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass: Mitigation; Speculative Store Bypass disabled v
ia prctl
  Spectre v1: Mitigation; usercopy/swaps barriers and __user
pointer sanitization
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB con
ditional; PBR SB-eIBRS SW sequence; BHI BHI_DIS_
S

```

```

S
Srbds:      Not affected
Tsa:        Not affected
Tsx_async abort: Not affected
Vmscape:    Not affected
Child process completed.
praneeth@PRANEETH:~$ ./1
Enter Linux command: lsblk
Parent PID : 1151
Child PID : 1152
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk 
sdb   8:16   0 159.4M  1 disk 
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0     1T   0 disk /mnt/wslg/distro

```

```

Child process completed.
praneeth@PRANEETH:~$ ./1
Enter Linux command: top
Parent PID : 1162
Child PID : 1163
top - 00:02:06 up 29 min, 1 user, load average: 0.07, 0.04, 0.00
Tasks: 28 total, 1 running, 27 sleeping, 0 stopped, 0 zombie
%Cpu(s):  0.2 us,  0.2 sy,  0.0 ni, 99.6 id,  0.0 wa,  0.0 hi,  0.1 si,  0.0 st
MiB Mem : 7756.6 total, 6598.9 free, 1036.0 used, 275.4 buff/cache
MiB Swap: 2048.0 total, 2048.0 free,  0.0 used, 6720.7 avail Mem

```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
283	mysql	20	0	983544	526140	44868	S	1.7	6.6	0:23.06	mysqld
1163	praneeth	20	0	8284	6112	3944	R	0.3	0.1	0:00.05	top
1	root	20	0	24252	15352	11540	S	0.0	0.2	0:01.25	systemd
2	root	20	0	3188	2208	2072	S	0.0	0.0	0:00.04	init-systemd(Ub
6	root	20	0	3188	2044	1940	S	0.0	0.0	0:00.00	init
46	root	19	-1	50412	16940	15632	S	0.0	0.2	0:00.22	systemd-journal
78	systemd+	20	0	22416	14396	11904	S	0.0	0.2	0:00.06	systemd-resolve
87	root	20	0	35368	12328	9232	S	0.0	0.2	0:00.19	systemd-udev
162	root	20	0	2888	1952	1820	S	0.0	0.0	0:00.03	chronyd-starter
163	root	20	0	4472	3116	2868	S	0.0	0.0	0:00.00	cron
164	message+	20	0	8792	5444	4716	S	0.0	0.1	0:00.09	dbus-daemon
172	root	20	0	44896	29868	14728	S	0.0	0.4	0:00.21	networkd-dispat
180	root	20	0	18436	9396	8152	S	0.0	0.1	0:00.10	systemd-logind
193	syslog	20	0	220548	5920	4628	S	0.0	0.1	0:00.11	rsyslogd
219	root	20	0	5402	2720	2536	S	0.0	0.0	0:00.01	agetty
247	_chrony	20	0	20424	7012	6212	S	0.0	0.1	0:00.07	chronyd
265	root	20	0	123460	32704	18088	S	0.0	0.4	0:00.17	unattended-upgr
267	_chrony	20	0	12096	2428	1788	S	0.0	0.0	0:00.00	chronyd
474	root	20	0	3184	1112	980	S	0.0	0.0	0:00.00	SessionLeader
475	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.18	Relay(480)
480	praneeth	20	0	6268	5536	3864	S	0.0	0.1	0:00.13	bash
481	root	20	0	8648	5532	4608	S	0.0	0.1	0:00.00	login
526	praneeth	20	0	22504	13392	10956	S	0.0	0.2	0:00.17	systemd
528	praneeth	20	0	22912	4120	2120	S	0.0	0.1	0:00.00	(sd-pam)
558	praneeth	20	0	6136	5416	3864	S	0.0	0.1	0:00.05	bash

901	root	20	0	1792300	12100	9444	S	0.0	0.2	0:00.14	wsl-pro-service
1017	polkitd	20	0	306444	8440	7624	S	0.0	0.1	0:00.13	polkitd
1162	praneeth	20	0	2768	1880	1784	S	0.0	0.0	0:00.00	1
1163	praneeth	20	0	8284	6112	3944	R	0.0	0.1	0:00.05	top

EXP 2:-

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>

int main()
{
    int src, dest;
    char buffer[1024];
    int bytes;

    src = open("input.txt", O_RDONLY);
    if(src < 0)
    {
        printf("Cannot open source file\n");
        return 1;
    }

    dest = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    while((bytes = read(src, buffer, sizeof(buffer))) > 0)
    {
        write(dest, buffer, bytes);
    }
    close(src);
    close(dest);
    printf("File copied successfully.\n");
    return 0;
}
```

OUTPUT :-

```
praneeth@PRANEETH:~$ nano 2.c
praneeth@PRANEETH:~$ gcc 2.c -o 2
praneeth@PRANEETH:~$ ./2
Cannot open source file
praneeth@PRANEETH:~$ |
```